

Program 5

Demonstrate CI Integration by Running Jenkins in a Docker Container

The program focuses on demonstrating Continuous Integration(CI) by deploying and running Jenkins inside a Docker container. This setup showcases how Docker simplifies Jenkins installation and enables automated build and integration workflows.

Project Structure:

program_5/

—>Dockerfile

—>docker-compose.yml

Procedure:

Step 1: Create a **Dockerfile** without any extensions and add the following content

```
nano Dockerfile
GNU nano 7.2 Dockerfile
#Use jenkins base
FROM jenkins/jenkins:lts

#Switch to root user to install tools
USER root

#Install Docker inside jenkins container
RUN apt-get update && apt-get install -y docker.io

#Use jenkins as user
USER jenkins

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

Step 2: Create a docker-compose.yml file with following code

nano docker-compose.yml

```
GNU nano 7.2                                docker-compose.yml
version: '3.8'
services:
  jenkins:
    container_name:jenkins-ci
    build: .
    ports:
      - "8080:8080"
      - "5000:5000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock
volumes:
  jenkins_home:
```

GNU nano 7.2 docker-compose.yml

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Step 3: Build an image and Run the container

docker build -t program_5 .

docker run -p 8080:8080 program_5

Step 4: Verify localhost details on browser

<http://localhost:8080>

